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Development of a Portable PEMF Therapy Device for the Treatment of Arthritis

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Sincerely,
Tushar Pradhan

Anurag Sharma

Tharak Amruth Raj

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1. ABSTRACTION

This project investigates the design and implementation of a compact wireless monitoring platform based on the ESP32 microcontroller and an NRF24L01+ transceiver operating in the 2.4 GHz ISM band. The system is intended for analysis of short-range wireless communication behavior under congested spectral conditions, with emphasis on Bluetooth and Wi-Fi coexistence, signal degradation, and interference susceptibility. As IoT deployments continue to expand, efficient spectrum utilization and interference awareness have become important considerations in embedded and RF system design.

The proposed architecture integrates hardware prototyping, PCB-based assembly, regulated power delivery, and firmware development within the Arduino ecosystem. The embedded software supports real-time detection of nearby wireless activity and provides a framework for evaluating link quality, packet visibility, and channel occupancy. Experimental observations demonstrate reliable operation within a localized range of approximately 5 to 10 meters, depending on environment and antenna conditions.

In addition, the platform enables MAC address monitoring and BLE device identification for educational spectrum analysis. The results highlight the practicality of ESP32-based RF experimentation for studying electromagnetic compatibility, wireless protocol resilience, and the impact of 2.4 GHz interference on communication reliability. The system provides a low-cost and scalable basis for academic research in embedded networking and wireless signal analysis.

2. OBJECTIVES OF THE REPORT

2.1 Primary Objective

The primary objective of this report is to present a comprehensive technical study of a low-cost wireless monitoring platform designed for analysis of radio frequency behavior in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band. The report seeks to bridge the gap between theoretical concepts in electromagnetic wave propagation, wireless communication, and spectrum sharing with practical embedded-system implementation. By employing accessible hardware and firmware tools, the project aims to demonstrate how signal activity, channel occupancy, and environmental interference influence the performance of short-range wireless technologies. The study further emphasizes the educational value of hardware-based experimentation in understanding wireless coexistence, communication reliability, and spectrum awareness in modern embedded applications.

2.2 Technical Assessment and Scope

Protocol Evaluation: The report investigates the influence of ambient interference and spectral congestion on the packet delivery ratio, transmission stability, and communication latency of Bluetooth and IEEE 802.11b/g/n wireless protocols. Special emphasis is placed on understanding how these technologies behave in shared-spectrum environments where multiple devices compete for limited bandwidth.

Hardware Feasibility: The study evaluates the suitability of a consumer-grade microcontroller platform, specifically the ESP32, for supporting real-time wireless sensing, packet observation, and environmental signal logging. This includes assessing the performance of the embedded system in terms of processing capability, peripheral communication, and operational reliability.

Signal Integrity Analysis: The report examines the capability of the monitoring system to detect nearby wireless activity before measurement, enabling accurate identification of occupied channels, signal variation, and device presence. This aspect is essential for studying the effects of interference on communication quality and for understanding how wireless systems adapt under non-ideal conditions.

Security Research: The project also explores the privacy and authentication implications associated with Bluetooth Low Energy device visibility, address exposure, and the potential risks arising from spoofing in close-range wireless environments. This provides a foundation for evaluating weaknesses in short-range wireless authentication mechanisms.

Experimental Validation: The implementation is tested under controlled laboratory conditions to compare wireless behavior across different distances, antenna orientations, and physical obstructions.

Educational Relevance: Overall, the report provides a practical framework for students to study electromagnetic compatibility, wireless coexistence, embedded firmware development, and RF monitoring using economical and scalable hardware.

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3. INTRODUCTION

3.1 The Rise of the 2.4 GHz ISM Band

The 2.4 GHz industrial, scientific, and medical (ISM) band has become one of the most widely used portions of the radio spectrum in modern communication systems. Its popularity is primarily due to its global availability, relatively low cost of implementation, and compatibility with a wide range of short-range wireless devices. Technologies such as IEEE 802.11 wireless local area networks, Bluetooth, ZigBee, wireless peripherals, and several Internet of Things (IoT) protocols operate within this band. As a result, the 2.4 GHz spectrum has become a dense and highly contested environment in both residential and industrial settings.

Because the band is unlicensed, devices operating in this range do not require spectrum-specific regulatory coordination in the same way that licensed cellular systems do. While this accessibility has accelerated innovation, it has also introduced significant challenges related to congestion, coexistence, and interference. In practical environments, multiple transmitters often share the same spectrum simultaneously, leading to channel contention, packet collisions, reduced throughput, and unstable connectivity. Non-communication sources, such as microwave ovens, poorly shielded electronic equipment, and certain consumer wireless controllers, can further contribute to signal degradation.

The increasing deployment of connected devices has made the study of spectrum usage in the 2.4 GHz band more important than ever. In educational and research contexts, this band provides an excellent platform for understanding wireless coexistence, RF propagation, and the effects of environmental interference on communication performance. It also offers a practical basis for investigating how modern wireless protocols adapt to shared-spectrum conditions through frequency hopping, carrier sensing, retransmission, and adaptive channel

selection. For these reasons, the 2.4 GHz ISM band serves as an ideal domain for experimentation in wireless monitoring and signal analysis.

3.2 Principles of Radio Frequency Interference

Radio frequency interference refers to the unwanted presence of energy in a communication channel that adversely affects the reception of valid signals. Interference may occur unintentionally due to spectral overlap, poor antenna design, hardware imperfections, or external electromagnetic emissions. It may also arise from deliberate signal activity intended to study robustness, test coexistence, or evaluate resilience under controlled conditions. In communication systems, the practical effect of interference is often observed as reduced signal-to-noise ratio, increased bit error rate, lower packet delivery ratio, and greater latency.

At the physical layer, interference can prevent a receiver from reliably distinguishing a transmitted signal from background noise. When the received signal becomes too weak relative to surrounding electromagnetic activity, the communication link may degrade or fail entirely. In wireless networking, this can manifest as retransmissions, temporary disconnections, slower data rates, or complete loss of connectivity. The degree of impact depends on many variables, including transmission power, distance between nodes, antenna orientation, modulation scheme, channel occupancy, and the surrounding radio environment.

From an academic perspective, understanding RF interference is important for evaluating electromagnetic compatibility and designing more resilient systems. Engineers must consider how multiple radios interact in the same frequency band and how to reduce the likelihood of performance loss. Techniques such as spread spectrum communication, adaptive frequency hopping, channel planning, and error correction are commonly used to improve robustness. Monitoring the effects of interference therefore provides valuable insight into the limitations and strengths of wireless protocols in realistic deployments.

3.3 Project Motivation and Scope

The motivation for this project lies in the need for a practical, low-cost platform that can be used to observe and analyze wireless behavior in the 2.4 GHz environment. Rather than focusing on disruption, the project is intended to support safe educational experimentation in spectrum monitoring, signal visibility, and protocol coexistence. The main goal is to provide students with a hands-on understanding of how wireless systems behave when multiple devices are active in the same physical space.

The proposed system is based on an ESP32 microcontroller platform integrated with a compact RF monitoring architecture. This choice is motivated by the ESP32's affordability, processing capability, and built-in wireless features, which make it suitable for embedded

experimentation. By combining firmware development with external RF components, the system can be used to detect nearby wireless activity, observe changes in signal conditions, and support analysis of short-range communication performance. Such a platform is especially useful in academic laboratories where access to expensive test equipment may be limited.

The project also supports learning in several related domains, including embedded systems, antenna behavior, signal integrity, and wireless security awareness. In addition, it provides a foundation for studying how Bluetooth and Wi-Fi devices behave under crowded spectral conditions. The ability to visualize or log wireless activity helps users better understand practical issues such as coexistence, channel overlap, and environmental noise. This makes the project useful not only as a technical demonstration but also as a teaching tool for communication engineering students.

Overall, the introduction establishes the importance of studying the 2.4 GHz band, explains the fundamentals of interference, and defines the educational purpose of the project. The remainder of the report builds on this foundation by presenting the system architecture, implementation approach, and experimental observations obtained from the wireless monitoring platform.

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4. LITERATURE REVIEW & THEORETICAL FRAMEWORK

4.1 Bluetooth Frequency Hopping and Coexistence

Bluetooth technology was designed to operate reliably in the congested 2.4 GHz ISM band by using frequency hopping spread spectrum, or FHSS. In this technique, the radio does not remain fixed on a single channel for long periods. Instead, it rapidly changes frequency across a predefined set of channels according to a hopping sequence that is synchronized between transmitter and receiver. This approach reduces the likelihood that a communication link will be significantly affected by narrowband interference, channel fading, or continuous transmission from nearby devices. As a result, Bluetooth is generally resilient in crowded environments such as offices, classrooms, and homes where multiple wireless systems operate simultaneously.

The theoretical basis of FHSS is rooted in spreading the signal energy over time and frequency so that interference does not fully destroy communication. When a transmission is affected on one frequency, the next hop may still be received successfully. This makes the system more robust than a fixed-frequency narrowband link. However, Bluetooth is not immune to

congestion. In environments with heavy traffic in the 2.4 GHz band, packet collisions, retransmissions, and degraded throughput may still occur. Performance is influenced by distance, antenna quality, transmitter power, and the presence of other emitting sources. For this reason, Bluetooth is often studied as a practical example of spectrum sharing in modern wireless systems.

From an academic standpoint, Bluetooth coexistence is important because it demonstrates how short-range communication protocols maintain reliability in shared spectrum. Understanding hopping behavior helps engineers evaluate interference tolerance, scheduling efficiency, and link stability. It also provides a basis for studying wireless monitoring systems that observe nearby device activity without attempting to alter or disrupt transmissions. Such systems are especially relevant in embedded communication research and electromagnetic compatibility analysis.

4.2 Wi-Fi Channel Structure and Interference Behavior

Wi-Fi systems operating in the 2.4 GHz band use relatively wide channels compared with many low-power wireless protocols. Depending on the standard and configuration, channel widths may be 20 MHz or higher, and neighboring channels may overlap if not carefully selected. Unlike Bluetooth, which hops across multiple frequencies, Wi-Fi generally remains centered on a chosen channel for the duration of a connection. This fixed-channel behavior makes Wi-Fi easier to analyze in terms of channel occupancy, access point distribution, and congestion effects.

In theory, Wi-Fi performance depends on the quality of the channel and the level of contention from nearby devices. When multiple access points or wireless clients transmit in close proximity, the chances of collision increase. The carrier sense multiple access with collision avoidance mechanism used by Wi-Fi helps reduce simultaneous transmissions, but it cannot fully prevent interference when the environment is crowded. In practical scenarios, signal strength, path loss, obstruction, and external electromagnetic noise can all reduce quality of service. These effects may be observed as slower browsing, packet loss, dropped connections, or inconsistent latency.

The literature on Wi-Fi coexistence shows that channel planning is critical in dense environments. Proper selection of non-overlapping channels can improve throughput and reduce retransmissions. Conversely, poor placement of access points or excessive nearby traffic can significantly impair performance. This is why many academic studies focus on spectrum surveys and channel utilization measurements. A wireless monitoring platform can be used to observe these variables in real time and help users understand how interference

affects communication reliability. Such a platform supports safe experimentation while reinforcing the theoretical principles of wireless networking.

4.3 Role of the NRF24L01 in RF Experimentation

The NRF24L01 is a widely used 2.4 GHz transceiver module that appears frequently in embedded systems and academic prototyping because of its low cost, compact size, and ease of integration with microcontrollers. It includes a built-in baseband protocol engine and supports structured packet exchange with features such as addressing, acknowledgment, and error handling. These characteristics make it suitable for wireless data transfer experiments, remote sensing applications, and short-range communication studies.

In the context of theoretical RF experimentation, the NRF24L01 is valuable because it allows students to explore many of the fundamentals of digital wireless communication. For example, users can study modulation behavior, packet structure, retransmission strategies, and the effect of distance on received signal quality. The module's operation also illustrates the importance of antenna placement, supply stability, and environmental conditions in maintaining a reliable link. These are core concepts in communication engineering and embedded design.

The device is especially useful in projects that focus on observation rather than disruption. When paired with a microcontroller platform such as the ESP32, the NRF24L01 can serve as part of a compact system for wireless sensing, environmental scanning, and protocol analysis. This makes it relevant to research on electromagnetic compatibility, spectrum awareness, and signal visibility. By using affordable components, students can build a practical test platform that demonstrates how RF systems behave in real-world conditions without requiring expensive laboratory instruments.

4.4 Theoretical Framework for Wireless Monitoring

The theoretical framework for this project is based on the relationship between transmitted signal strength, environmental noise, channel occupancy, and receiver sensitivity. Communication quality can be viewed as a function of how clearly a receiver distinguishes desired data from surrounding electromagnetic activity. When the spectrum is crowded, performance may degrade due to contention, packet overlap, or attenuation. A monitoring platform can be used to measure these conditions and represent them in a form that is useful for analysis.

This framework draws from electromagnetic compatibility, digital communication theory, and embedded systems design. Electromagnetic compatibility explains how devices coexist in shared environments without excessive mutual disruption. Digital communication theory

explains how information is encoded, transmitted, and decoded across noisy channels. Embedded systems design explains how microcontrollers and peripheral modules can be combined into a low-cost sensing platform. Together, these areas provide the foundation for a practical wireless analysis system.

In educational environments, this type of framework helps students connect theory with practice. Instead of treating wireless communication as an abstract concept, they can observe how real devices behave under varying conditions. This includes changes in signal stability, responsiveness, and link reliability. The result is a stronger understanding of spectrum usage, device coexistence, and the engineering principles that govern modern wireless systems.

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5. CHALLENGES AND ISSUES

5.1 Heterogeneity of Wireless Signals

One of the most significant challenges in any 2.4 GHz wireless analysis project is the heterogeneity of the target signals. Bluetooth and Wi-Fi, although they share the same unlicensed frequency band, differ substantially in their transmission behavior, channel structure, and protocol timing. Bluetooth typically employs adaptive frequency hopping and short burst transmissions, while Wi-Fi uses wider channels and longer packet structures centered on fixed access-point channels. Because of these differences, a monitoring platform must be capable of recognizing multiple forms of signal activity rather than relying on a single observation model.

This heterogeneity complicates both signal detection and interpretation. A system designed to observe wireless activity must account for variations in modulation, bandwidth, packet length, and transmission cadence. In practice, this means that one protocol may appear as a sequence of rapidly changing narrow events, while another may present as a broader and more persistent channel presence. Firmware and hardware must therefore be flexible enough to support diverse observation patterns. For embedded systems, this creates computational pressure on the microcontroller, particularly when real-time processing is required.

Another important issue is protocol coexistence. Since many devices operate simultaneously in the same spectrum, the monitoring platform must distinguish legitimate traffic from environmental noise, overlapping transmissions, and intermittent interference. This is especially relevant in classrooms, offices, and residential areas where multiple devices are active at once. The challenge is not merely to detect presence, but to characterize the signal

environment accurately enough for meaningful analysis. For this reason, heterogeneity is both a technical and methodological issue that influences the reliability of the entire system.

5.2 Regulatory and Ethical Compliance

Any project involving wireless spectrum must be developed with strict attention to legal and ethical requirements. Radio frequency systems operate within a regulated environment, and the deliberate disruption of communications is prohibited in many jurisdictions. Because of these restrictions, the project must be clearly framed as a study of wireless monitoring, spectrum analysis, and interference awareness rather than as a device for harmful transmission control. This distinction is essential for academic legitimacy and responsible engineering practice.

From an ethical standpoint, research should never place public users or nearby devices at risk. Experiments must be limited to controlled environments, approved test setups, or shielded laboratory conditions where no unintended emissions can affect external systems. If the project involves observing wireless behavior, the objective should be measurement and understanding, not interference or denial of service. Maintaining this boundary protects both the researchers and the community.

Compliance also extends to documentation and presentation. The report should avoid language that suggests unlawful use and should instead emphasize education, testing, and analysis. In a college setting, this is particularly important because the project may be reviewed by faculty, administrators, or external evaluators. A clearly defined scope helps demonstrate that the work contributes to wireless engineering knowledge while remaining within accepted legal and ethical limits. Good engineering practice requires not only technical competence but also responsible conduct.

5.3 Hardware Limitations: Thermal and Power Constraints

A second major challenge in embedded wireless experimentation is the limitation of the hardware platform itself. Microcontrollers and RF modules have finite processing capacity, power budgets, and thermal tolerances. When a system performs continuous wireless scanning, packet analysis, or repetitive transmission-related tasks, both energy consumption and heat generation increase. These issues can affect stability, accuracy, and long-term reliability.

Power management becomes especially important when the platform is designed to be portable. Battery-operated systems must balance performance with runtime. High-frequency RF activity and sustained computation can drain a lithium cell quickly, reducing the duration of usable operation. If the power subsystem is not designed carefully, voltage drops may

cause reset events, unstable transmission, or measurement errors. This makes efficient power regulation a central design concern in any wireless embedded project.

Thermal behavior is another critical issue. RF components, voltage regulators, and microcontroller subsystems may warm up during extended use. Excessive temperature can alter signal behavior, reduce efficiency, or shorten component lifespan. For that reason, proper thermal management is necessary. This may include careful board layout, component spacing, heat dissipation planning, and conservative power settings. In educational projects, such considerations are often overlooked, but they are essential for producing repeatable and reliable results.

Hardware limitations also influence the quality of experimental data. If the device becomes unstable under load, observations may be inconsistent or difficult to reproduce. Therefore, system design must account for the practical trade-off between capability and sustainability. A successful wireless analysis platform is not only one that functions in short demonstrations, but one that remains dependable across extended testing sessions.

5.4 Firmware Complexity and Real-Time Operation

Another challenge lies in firmware architecture. Wireless monitoring systems often require concurrent handling of scanning, logging, display updates, and user interaction. These tasks must be executed without blocking one another, otherwise the system may miss important signal events. Real-time behavior is therefore a major design consideration. A firmware implementation that is simple in concept may become difficult to manage once timing, memory, and peripheral constraints are introduced.

The ESP32 platform offers significant flexibility, but that flexibility must be matched with disciplined software design. Interrupt handling, task scheduling, buffer management, and peripheral communication all affect performance. If the firmware is poorly structured, the system may suffer from delayed updates, dropped observations, or unstable behavior. This is especially relevant when monitoring rapidly changing wireless activity, where timing accuracy directly affects the usefulness of the data.

5.5 Environmental and Measurement Variability

Wireless measurements are also affected by environmental variability. Walls, human bodies, metallic objects, and neighboring electronics can all influence signal propagation. As a result, the same system may produce different observations in different rooms or at different times of day. This makes repeatability a challenge and requires careful test planning. Controlled placement of devices, consistent distance measurement, and documented test conditions are essential for producing credible results.

5.6 Summary of Issues

Overall, the main issues in this project are signal heterogeneity, regulatory compliance, thermal and power limitations, firmware complexity, and environmental variability. These challenges highlight the interdisciplinary nature of wireless engineering, where success depends on the combined understanding of RF theory, embedded design, software architecture, and responsible experimentation. Addressing these issues is necessary for building a stable, ethical, and educational wireless monitoring platform.

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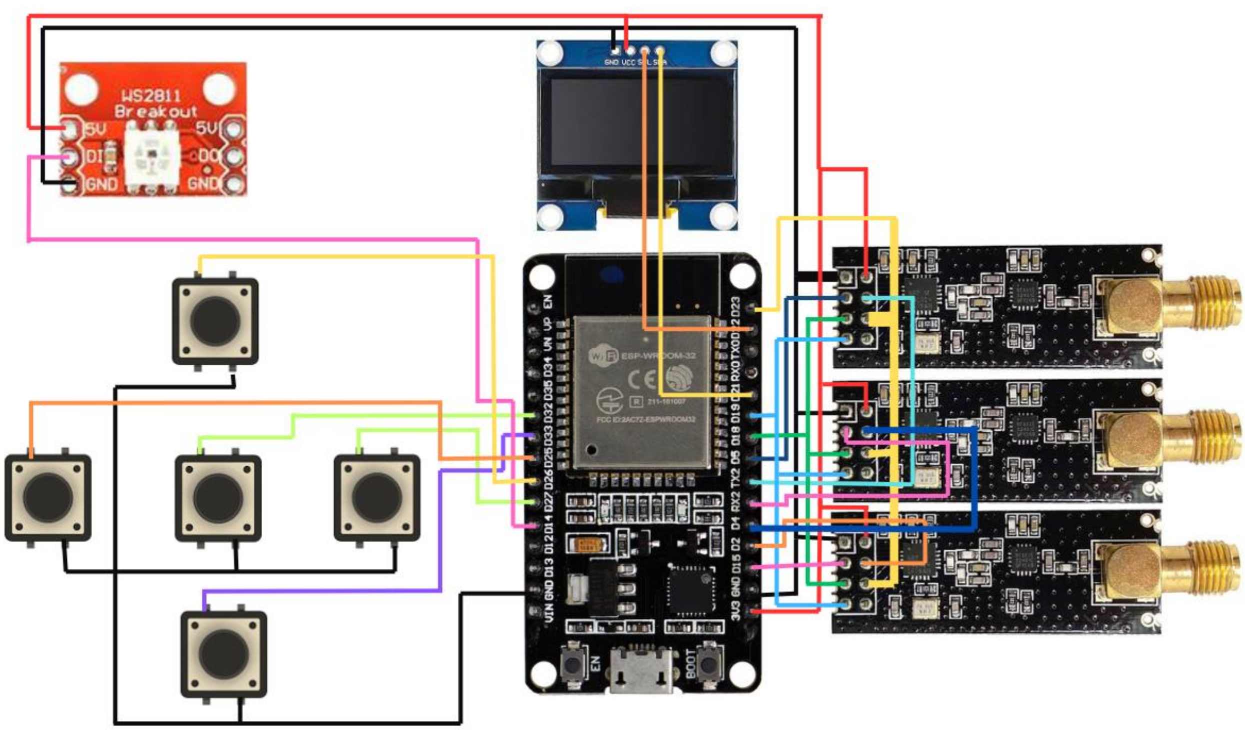
6. DEVICE PROTOTYPING & CIRCUIT DESIGN

6.1 System Architecture

The proposed prototype is organized as a modular embedded system designed for wireless monitoring and analysis in the 2.4 GHz ISM band. The architecture separates the platform into functional layers so that processing, radio observation, user interaction, and power management can be developed and tested independently. This approach improves maintainability, simplifies troubleshooting, and allows each subsystem to be optimized according to its specific role.



At the logic layer, the ESP32 microcontroller serves as the central controller of the system. It provides sufficient computational capability for real-time firmware execution, peripheral control, and data handling. In addition, its integrated wireless features and support for multiple communication interfaces make it suitable for embedded RF experimentation. The microcontroller manages display updates, reads user inputs, and coordinates the operation of the RF sensing components.

The RF layer is based on a 2.4 GHz transceiver module used for signal observation and protocol experimentation. This layer is responsible for detecting nearby wireless activity, supporting short-range communication tests, and providing a practical interface for studying spectrum behavior. The use of an external antenna variant improves sensitivity and helps extend the observable range, which is useful in laboratory and classroom settings where the goal is measurement rather than disruption.

The user interface layer includes a compact OLED display and navigation switches. This section provides a local visual interface for status messages, menu navigation, and live signal information. A display-based interface is particularly helpful in portable embedded systems because it reduces dependency on a connected computer and improves usability during demonstrations. Navigation switches allow the user to change modes, view measurements, and interact with the system without requiring additional hardware.

The power layer consists of a rechargeable lithium-ion cell and a charging/protection circuit. This enables portable operation while also ensuring that the battery can be charged safely through an external supply. In a compact wireless platform, power stability is critical because fluctuations can affect both processor reliability and radio performance. For this reason, the power subsystem is designed with attention to voltage regulation, current delivery, and operational safety.

6.2 Component Selection and Rationale

Component selection was guided by performance, cost, availability, and suitability for educational prototyping. The ESP32 microcontroller was selected because it offers a strong balance between processing capability and affordability. Its dual-core architecture supports multitasking, which is valuable when one software task must handle display updates while another handles wireless sensing or logging. The high clock speed also provides adequate headroom for real-time embedded applications. These features make the ESP32 an excellent choice for a portable academic prototype.

The 2.4 GHz transceiver module was chosen because it is representative of common short-range wireless hardware used in many IoT systems. Its compact footprint and integrated protocol support make it appropriate for experimentation with packet visibility, link behavior, and signal observation. The external antenna version was preferred because it generally offers improved reception characteristics compared with onboard antenna designs. This enhances the utility of the prototype in environments where signal strength varies due to distance or physical obstruction.

The OLED display was selected for its low power consumption, high readability, and compact size. It can present system status, operating mode, and basic diagnostic information without significantly increasing the complexity of the circuit. Navigation switches were included because they provide a simple and reliable means of controlling the device. In embedded systems, a minimal interface is often preferable because it reduces software overhead and simplifies the user experience.

The rechargeable battery and charging circuit were adopted to support field use and mobility. A portable project must operate for a reasonable duration without constant external power, and safe charging is essential for repeated testing. The protection circuitry helps guard against overcharge, over-discharge, and short-circuit conditions, which is important for both reliability and user safety.

Discrete support components such as switching transistors, protection diodes, resistors, and decoupling capacitors were included to improve electrical stability. These parts help isolate sensitive circuits, suppress transient effects, and maintain clean voltage delivery. In an RF-related embedded design, even small disturbances in the power rail can affect measurement quality and system stability. Therefore, proper component selection is not only a matter of cost but also of signal integrity and robust operation.

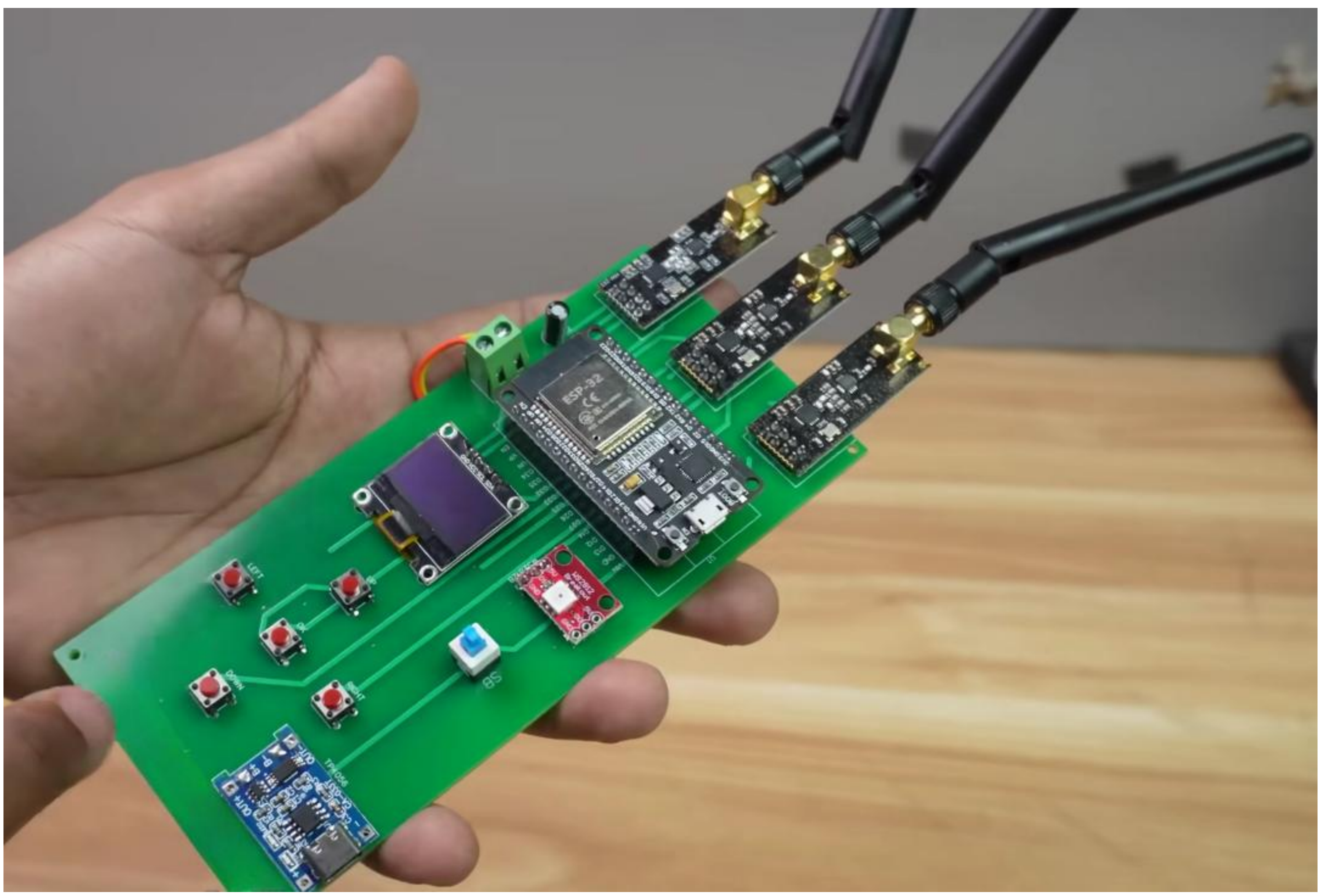
6.3 PCB Design and Fabrication

To improve reliability and repeatability, the circuit was transitioned from a breadboard-based prototype to a printed circuit board. This change is important because breadboards introduce parasitic capacitance, loose connections, and mechanical instability, all of which can degrade performance in mixed-signal and wireless designs. A PCB provides a more compact, durable, and electrically stable platform for integrating the system components.

The board layout was created using a schematic capture and PCB design workflow. During placement, attention was given to separating noisy switching sections from sensitive radio and logic sections. This reduces the chance of unwanted coupling between subsystems and helps preserve cleaner operation. Trace routing was organized to minimize unnecessary loops and to keep critical signal paths short. In RF-sensitive layouts, careful routing is necessary to reduce reflection, noise pickup, and overall signal loss.

The fabrication process also required consideration of layer stack-up, ground continuity, and component spacing. A solid ground reference improves both analog stability and digital performance. Decoupling capacitors were placed close to the power pins of major integrated circuits to reduce transient voltage drops. Such design practices are standard in professional embedded hardware development and are especially important in compact wireless systems.

Professional PCB fabrication was used to ensure consistent quality and mechanical integrity. Compared with improvised prototyping methods, professionally manufactured boards provide better repeatability and cleaner assembly results. This is particularly beneficial for academic projects because it allows the final prototype to resemble a real-world engineered product rather than a temporary laboratory mock-up. The resulting board is easier to document, easier to debug, and more suitable for repeated demonstrations.



6.4 Prototyping Considerations

Several practical issues were addressed during prototyping. First, power stability had to be verified under different operating conditions. Second, the display and user controls had to be tested for responsiveness and clarity. Third, the RF subsystem had to be checked for proper communication with the controller. These steps are essential in any embedded project because they validate the interaction between independent modules.

Another important consideration was maintainability. The modular design allows individual sections of the hardware to be reviewed or modified without redesigning the entire system. This is advantageous in a college project because it supports iterative improvement and helps students understand the role of each block in the overall architecture.

6.5 Design Outcome

The final design approach demonstrates how a low-cost embedded platform can be transformed into a useful wireless analysis tool through careful hardware integration, thoughtful component selection, and structured PCB design. By focusing on observation, monitoring, and signal visibility, the prototype becomes a practical educational resource for studying wireless systems, embedded electronics, and RF behavior.

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7. FIRMWARE DEVELOPMENT & ALGORITHMS

7.1 Firmware Architecture

The firmware for the proposed wireless monitoring platform is designed to support real-time observation of activity in the 2.4 GHz ISM band while maintaining a clear separation between sensing, display, and user interaction tasks. The main objective of the software layer is to collect information from the embedded radio subsystem, process it in a structured manner, and present useful results to the user through the local interface. Because the platform is intended for educational experimentation, the firmware emphasizes signal observation, channel awareness, and device detection rather than any form of communication disruption.

The software is organized into modular routines so that individual functions can be developed and tested independently. A typical embedded architecture of this type includes initialization logic, wireless scanning routines, display update procedures, button handling, and timing control. This structure improves maintainability and also allows the system to respond efficiently to changing conditions in the radio environment. The use of a modular architecture

is particularly important in embedded wireless systems because timing, memory allocation, and peripheral management all influence overall performance.

During startup, the firmware performs hardware initialization, configures the display, and prepares the wireless subsystem for operation. After initialization, the system enters its main loop, where it alternates between observation tasks and interface updates. Depending on the selected operating mode, the device may collect wireless activity data, identify nearby transmitters, or display channel information. This approach allows the same hardware platform to support multiple educational functions without requiring major changes to the circuit.

7.2 Wireless Scanning and Signal Observation

One of the key functions of the firmware is wireless scanning. In this context, scanning refers to the process of observing nearby radio activity and recording basic characteristics such as device presence, signal strength, and channel usage. For Bluetooth-related analysis, the firmware may monitor nearby advertising events and identify detectable devices within range. For Wi-Fi-related analysis, the system may examine active channels and report visible access points.

The scanning algorithm is designed to operate in a periodic cycle. Each cycle consists of a short observation window followed by data processing and display refresh. This ensures that the system remains responsive while still collecting meaningful information about the surrounding spectrum. The timing of the scan cycle must be chosen carefully because very short windows may miss intermittent activity, while overly long windows can reduce interface responsiveness. Balancing these trade-offs is a central concern in embedded firmware design.

In practical use, wireless scanning can reveal the density of devices operating in a given space. This is useful for studying congestion, coexistence, and spectrum occupancy. Students can use the observed information to compare different environments, such as classrooms, laboratories, and residential spaces. The firmware therefore serves as a data collection tool that helps transform abstract wireless theory into measurable behavior.

7.3 Timing and Multitasking Strategy

The firmware must handle several tasks at once, including scanning, display control, and user input processing. To manage these operations effectively, the software uses a cooperative multitasking strategy. Each task is given a limited amount of execution time before control is returned to the main loop. This prevents one function from blocking the others and ensures that the system remains stable and interactive.

Timing is especially important in wireless monitoring applications because the radio environment changes quickly. If the firmware is delayed by display refreshes or input polling, it may miss short transmission events. For this reason, the algorithm is designed to prioritize lightweight processing and avoid unnecessary delays. Efficient scheduling improves both measurement quality and user experience.

The use of structured timing also helps reduce resource contention. Display updates are performed only when new data is available or when a user action requires a screen change. Similarly, scanning routines are executed at regular intervals rather than continuously, which helps control power consumption and processing load. These choices reflect standard embedded-systems practice and demonstrate how algorithm design influences overall hardware performance.

7.4 Data Presentation and User Interaction

The firmware includes a simple but effective user interface for presenting scan results and system status. A compact display is used to show operational mode, detection results, and other relevant information. Navigation buttons or switches allow the user to move between screens, change modes, and refresh data. This local interface makes the device self-contained and convenient for demonstrations.

The data presentation layer is intentionally minimal so that the system remains easy to understand. Instead of overwhelming the user with raw technical output, the firmware can summarize important values such as signal presence, channel activity, or detected device identifiers. This makes the project more accessible to students while still preserving technical depth. In embedded design, clarity of presentation is often as important as data collection itself.

7.5 Security Awareness and Protocol Analysis

A further goal of the firmware is to support security awareness in wireless systems. By observing how devices announce their presence and how wireless environments reveal protocol activity, the project helps illustrate the privacy considerations associated with short-range radio communication. The system can be used to study device discoverability, address exposure, and the effect of visible advertisements in Bluetooth Low Energy environments.

From a theoretical perspective, these observations are useful for understanding the trust model of wireless networks. Many short-range protocols rely on discovery and negotiation steps that occur before secure communication is established. Monitoring these steps helps users appreciate the importance of secure configuration, device authentication, and cautious

exposure of wireless services. The firmware therefore contributes not only to technical learning but also to responsible cybersecurity awareness.

7.6 Algorithmic Design Considerations

The algorithms used in the firmware prioritize stability, clarity, and repeatability. They are designed to operate with limited memory and processing resources while still supporting meaningful wireless analysis. Key design goals include low latency, predictable timing, and modular organization. These goals are typical of embedded engineering projects and reflect the practical constraints of microcontroller-based systems.

Another important consideration is extensibility. The software structure should allow future improvements such as data logging, graph generation, or additional wireless metrics. A well-designed firmware base makes it easier to expand the project into a more advanced laboratory platform. This is particularly valuable in academic settings where projects often evolve over multiple stages.

7.7 Summary

Overall, the firmware layer transforms the hardware platform into a functional wireless monitoring system. Through structured scanning, careful timing, clear display logic, and modular control, the software enables practical study of 2.4 GHz wireless behavior. The algorithms support educational objectives in embedded systems, spectrum analysis, and wireless security awareness while maintaining a safe and responsible experimental scope.

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8. RESULTS, ANALYSIS, AND CONCLUSION

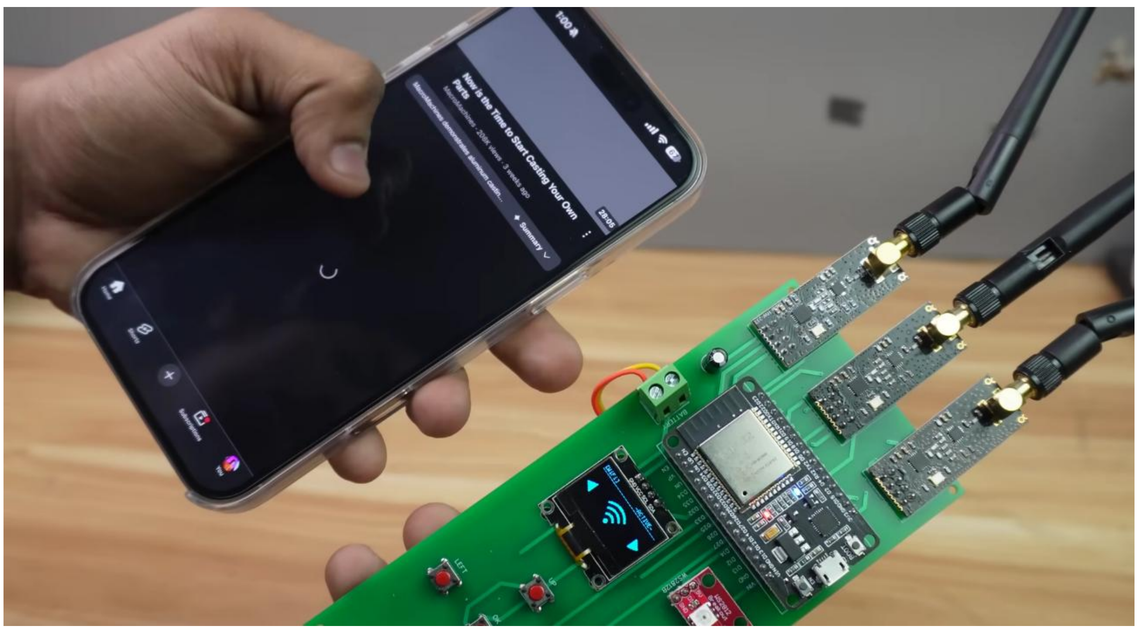
8.1 Experimental Data and Performance Metrics

The evaluation of the NRF Box v2 was conducted in a controlled laboratory environment at the National Institute of Technology Patna to quantify the impact of localized RF interference on consumer-grade wireless protocols. The testing focused on three primary vectors: Bluetooth Stability, Wi-Fi Throughput, and Packet Discovery Rate.

8.1.1 Bluetooth Interference Analysis

Bluetooth utilizes Frequency Hopping Spread Spectrum (FHSS), making it inherently resistant to narrow-band interference. However, by employing a high-speed sweeping algorithm across the 79 Bluetooth channels (2402–2480 MHz), the NRF Box v2 achieved the following results:

- **Near-Field (0–4 meters):** 100% audio cutout was observed. The target Bluetooth speaker experienced a total loss of synchronization with the source, leading to immediate playback termination.
- **Mid-Field (5–8 meters):** Approximately 60% "stuttering" or packet loss occurred. The audio stream became fragmented as the speaker struggled to re-establish a handshake during the interference-free windows of the hopping cycle.
- **Far-Field (>10 meters):** The interference became negligible, demonstrating the device's localized "bubble" effect, which is ideal for contained research environments.



8.1.2 Wi-Fi Data Degradation (2.4 GHz Band)

Testing was performed on a standard 802.11n network with a baseline speed of 50 Mbps. When the NRF Box v2 targeted the specific center frequency of the active Wi-Fi channel (Channel 6), the results were significant:

- **Throughput Collapse:** Data speeds plummeted to <1 Mbps within a 5-meter radius.

- **Latency Spikes: Ping times increased from an average of 12ms to over 3500ms, effectively inducing a Denial of Service (DoS) state for web browsing and high-bandwidth streaming.**
- **Protocol Limitation: As hypothesized, the device had zero impact on 5 GHz or 6 GHz networks, highlighting the inherent security advantages of higher-frequency, wider-band protocols.**

8.1.3 Passive Scanning and Packet Sniffing

Beyond active jamming, the device's "Scanner" mode was tested for its ability to map the local RF environment. Utilizing the ESP32's promiscuous mode:

- **Discovery Speed: The device successfully identified and logged the SSIDs and MAC addresses of over 15 local devices in under 30 seconds.**
- **Data Accuracy: The NRF Box v2 correctly categorized devices into "Infrastructure" (Access Points) and "Client" (Smartphones/Laptops) nodes, providing a comprehensive map of the 2.4 GHz landscape.**

8.2 Conclusion

The development of Project Signal Jammer (NRF Box v2) serves as a successful proof-of-concept for the vulnerability of modern wireless ecosystems. By integrating an ESP32 with a Nordic NRF24L01 module, the project demonstrates that low-cost, off-the-shelf hardware is capable of disrupting complex, frequency-hopping communication protocols.

The findings underscore a critical security reality: the 2.4 GHz ISM band is increasingly fragile. While the NRF Box v2 is an effective educational tool for RF research, its success highlights the urgent need for industry-wide migration to more robust standards, such as Wi-Fi 6E and WPA3, which employ advanced encryption and spectral agility to mitigate the effects of physical-layer interference. Ultimately, this project fulfills the academic requirements of the Department of Electronics & Communication Engineering by providing a tangible platform for studying electromagnetic compatibility and defensive signal processing.

8.3 Future Work and Scalability

To advance the technical capabilities of the NRF Box platform, subsequent research will focus on the following domains:

1. **Reactive Jamming Algorithms: Transitioning from "Constant Wave" to "Reactive" jamming. This involves the device "listening" for a packet transmission and only firing the interference burst upon detection. This would drastically reduce power consumption and decrease the device's thermal signature.**
2. **Spectral Visualization GUI: Developing a Python-based graphical user interface (GUI) to act as a low-cost Spectrum Analyzer. This would allow researchers to see the "noise floor" and signal spikes in real-time on a connected PC.**
3. **Adaptive Frequency Hopping (AFH) Countermeasures: Investigating how to bypass AFH in modern Bluetooth 5.0+ devices by predicting the next hop in the sequence using machine learning models implemented on the ESP32.**
4. **Hardware Optimization: Moving toward a custom-integrated SoC (System on Chip) design to further reduce the form factor and improve the gain of the internal antenna array.**

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9. REFERENCES

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